

Comparing Modern Control to Reinforcement Learning Control for Spacecraft Proximity Operations^{*}

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Abstract: In-Space or On-orbit Servicing, Assembly, and Manufacturing (ISAM/OSAM) is a growing field in developing space robotics to repair or refuel existing satellites, assemble large structures, and manufacture components in space. A foundational capability in this domain is the development of a control approach for OSAM vehicles to safely egress from the servicing or assembly area. This work compares the performance of model-based optimal control techniques to a Reinforcement Learning (RL) approach for the egress problem. Simulation results are presented where control efficiency is used as a primary metric for comparison.

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Keywords: Optimal Control; Aerospace; Machine Learning.

1. INTRODUCTION

In-space and On-Orbit Satellite Servicing, Assembly, and Manufacturing (ISAM/OSAM) are growing industries. ISAM covers all servicing, assembly, and manufacturing jobs off-Earth and OSAM is a subsection of ISAM solely for on-orbit purposes, with capabilities such as satellite repairs, maintenance, and part installations (Arney et al., 2021). Past OSAM missions include the on-orbit assembly of the International Space Station and multiple satellite servicing missions for the Hubble space telescope. Currently, unmanned spacecraft operate solely on human issued commands, but future missions will require the ability to make split second decisions. In these instances, autonomous and automatic control systems have the potential to maintain vehicle safety and mission assurance.

A fundamental capability of any OSAM/ISAM mission is an efficient approach into a servicing or assembly area (*ingress*) and an accompanying safe exit away from a servicing area (*egress*), while maintaining a safe distance from the chief to avoid collisions. An efficient feedback controller that can guarantee safety for the deputy satellite while

it circumnavigates the chief is, therefore, a crucial tool for egress purposes. To this end, this research compares model-based optimal control techniques, including Linear Quadratic Regulation (LQR), Linear Quadratic Integration (LQI), and Linear Quadratic Tracking (LQT), against Reinforcement Learning (RL) for the egress problem.

Previous related work includes utilizing a switching back-up controller to safely park a deputy spacecraft in an orbit about a chosen chief (Mote et al., 2021). Other investigations have utilized nonlinear state-dependent Riccati equation (SDRE) control technique and an LQT control technique to follow desired trajectories for translational maneuvers between spacecraft while also accounting for attitude (Lee and Pernicka, 2010). This paper investigates if there is a way to solve this problem without implementing a switching back-up controller.

The contributions of this paper are twofold: first, the development of LQR, LQI, LQT, and RL control solutions for safe egress from close proximity to a chief spacecraft to a distant elliptical natural motion trajectory (eNMT); and second, the comparison of performances with respect to time and largest cumulative variation in velocity.

This remainder of this paper is organized as follows: Sec. 2 introduces the problem statement, Sec. 3 illustrates an explanation of the control methods, while Sec. 4 shows the simulation results for four different cases of each control method. Finally, conclusions with a discussion of proposed future work are presented in Sec. 5.

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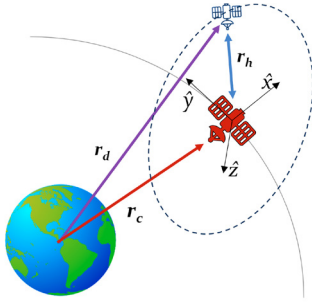


Fig. 1. Hill's Reference Frame.

2. PROBLEM FORMULATION

The OSAM technique utilized in this work is the egress of a deputy spacecraft from a chief spacecraft to a relative parking orbit, where it waits for ingress instructions for tasks such as repairs, maintenance, or refueling. The orbits used are described in the linearized Clohessy-Wiltshire equations, which are derived in Hill's reference frame.

2.1 Hill's Reference Frame

The Hill's reference frame, shown in Fig. 1, centers on the chief spacecraft, where $\hat{x}$ pointing outwards from Earth's center, $\hat{z}$ being aligned with the angular momentum vector of an orbit, and $\hat{y}$ complete an orthogonal coordinate system (for a circular orbit, $\hat{y}$ and the inertial orbital velocity chief are aligned). Additionally, $\mathbf{r}_h$ is the position vector pointing from the chief to the deputy, $\mathbf{r}_c$ is a vector from Earth's center to the chief, and $\mathbf{r}_d$ is the vector from Earth's center to the deputy.

2.2 The Clohessy-Wiltshire Equations

The Clohessy-Wiltshire (CW) equations are the linearized equations of motion used for a deputy satellite orbiting a chief satellite (Clohessy and Wiltshire, 1960):

$$\ddot{x} = 2n\dot{y} + 3n^2x + (F_x/m) \quad (1)$$

$$\ddot{y} = -2n\dot{x} + (F_y/m) \quad (2)$$

where n is the mean motion of the spacecraft [rad/s], and m is the mass of the deputy [kg]. For this problem, out-of-plane (z -axis) motion is neglected for simplicity. These equations can be written in state-space form as:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (3)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} \quad (4)$$

where the state vector is $\mathbf{x} = [x, y, \dot{x}, \dot{y}]^T \in \mathbb{R}^4$, the control vector is $\mathbf{u} = [F_x, F_y]^T \in [-u_{max}, u_{max}]^2$ and $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ matrices are derived from the CW equations converted to state space. The linearized CW equations are based on the following assumptions (Petersen et al., 2021):

- both satellites are rigid bodies,
- the mass of Earth $\gg$ the mass of the satellites,
- the mass loss of the spacecraft during maneuvers $\ll$ the total mass of the spacecraft,
- the chief spacecraft is in a circular orbit around Earth,
- the distance from the chief spacecraft to the deputy spacecraft ($\hat{r}_h$) $\ll$ the distance from the chief spacecraft to Earth's center ($\hat{r}_c$).

To guarantee the bounded relative orbit assigned to the deputy will not have linear drift, initial conditions (ICs) must be placed on $\dot{y}$. A similar IC is chosen for y to avoid any offset conditions. The orbit chosen for this paper is an in-plane closed eNMT (Alfriend et al., 2010), so the ICs set on the system are $y(0) = (2/n)\dot{x}(0)$ and $\dot{y}(0) = -2nx(0)$. Note for simulations in this paper, $x(0) = 1$ and $\dot{x}(0) = 0.001$. Commonly referred to as a *parking trajectory*, this is a stable orbit where once the directional thrust $\mathbf{u} = 0$, the deputy will stay on track in the orbit with no extra control needed (Mote et al., 2021). The shape of the eNMT ICs is given by

$$(x^2/a^2) + (y^2/b^2) = 1 \quad (5)$$

where $a = 200$ m is the length of the major radius, and $b = 100$ m is the length of the minor radius. The orbital rate of the chief satellite is $n = 0.001027$ and the satellite mass is $m = 12$ kg to symbolize a 6-unit CubeSat. When $\mathbf{u} = 0$, the CW equations are reduced to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$.

There are two control limits applied to the systems: a maximum relative velocity in any direction $[v_x, v_y]$ equal to 0.5 m/s, and a maximum value for the control inputs $[u_x, u_y]$ set to $[-u_{max}, u_{max}] = [-1, 1]$ N. Each simulation is limited to a maximum of two orbits, for a total chosen orbital period of 6,117 seconds.

3. CONTROL DESIGN

This section describes three methods from classical control theory namely, LQR, LQI, and LQT (Lewis and Syrmos, 1995) as well as a machine learning method, namely, RL.

3.1 Linear Quadratic Regulation (LQR)

LQR is an optimal state-feedback controller for a linear dynamical system that minimizes a quadratic cost on the states and control inputs to stabilize a closed-loop (CL) control system Lee and Pernicka (2010). The performance index to be minimized is:

$$J = \frac{1}{2} \int_0^\infty [\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}] dt \quad (6)$$

where $\mathbf{Q} \geq 0$ and $\mathbf{R} > 0$ are symmetric matrices. The transition state weight $\mathbf{Q}$ is a general diagonal matrix selected to provide appropriate weight to limit the states, and the control weight $\mathbf{R}$ is a diagonal matrix that can be weighted for the position states x and y independently to limit control. The optimal feedback control of the system is computed by solving the Algebraic Riccati Equation (ARE) and the time-invariant Kalman gain ($\mathbf{K}_\infty$). The resulting control law is:

$$\mathbf{u} = -\mathbf{K}_\infty(\mathbf{x} - \mathbf{r}) \quad (7)$$

where $\mathbf{r}$ is the reference trajectory which in our case takes the form $\mathbf{r} \in \mathbb{R}^4$. With the optimal control $\mathbf{u}$, $\dot{\mathbf{x}}$ can be found using Eq. 3.

3.2 Linear Quadratic Integration (LQI)

LQI is an extension of the standard LQR framework that includes integral action to the optimal control design to eliminate steady-state tracking or regulation errors. In our case, defining the variable $\boldsymbol{\xi}$:

$$\boldsymbol{\xi} = \int_0^T (\mathbf{r} - \mathbf{y}) dt \quad (8)$$

where the reference values $\mathbf{r} \in \mathbb{R}^2$, and combining Eq. 3 with Eq. 8, the augmented system equations become:

$$\begin{aligned} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\xi}} \end{bmatrix} &= \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\xi} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u} + \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix} \mathbf{r} \\ \dot{\bar{\mathbf{x}}} &= \bar{\mathbf{A}}\bar{\mathbf{x}} + \bar{\mathbf{B}}\bar{\mathbf{u}} + \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix} \mathbf{r} \end{aligned} \quad (9)$$

The performance index of LQI remains quadratic:

$$J = \frac{1}{2} \int_0^\infty [\bar{\mathbf{x}}(t)^T \bar{\mathbf{Q}} \bar{\mathbf{x}}(t) + \bar{\mathbf{u}}(t)^T \bar{\mathbf{R}} \bar{\mathbf{u}}(t)] dt.$$

The ARE and a time-variant Kalman gain are utilized to calculate the LQI control law,

$$\bar{\mathbf{u}} = -\bar{\mathbf{K}}(t)\bar{\mathbf{x}}. \quad (10)$$

The final solution is found with a closed-loop system version of Eq. 9.

3.3 Linear Quadratic Tracking (LQT)

Linear Quadratic Tracking (LQT) uses the linear dynamic system in state space form presented in Eq. 3 to minimize the performance index:

$$\begin{aligned} J(t_o) &= \frac{1}{2}(\mathbf{C}\mathbf{x}(T) - \mathbf{r}(T))^T \mathbf{P}(\mathbf{C}\mathbf{x}(T) - \mathbf{r}(T)) \\ &+ \frac{1}{2} \int_{t_o}^T [(\mathbf{C}\mathbf{x} - \mathbf{r})^T \mathbf{Q}(\mathbf{C}\mathbf{x} - \mathbf{r}) + \mathbf{u}^T \mathbf{R} \mathbf{u}] dt. \end{aligned} \quad (11)$$

The main difference between LQR and LQT is an additional auxiliary function $\mathbf{v}$. The previously used weights and a new final state weight $\mathbf{P}$ are all assigned to the model, as well as the final time $T \neq \infty$. The weight assumptions are $\mathbf{Q}, \mathbf{P} \geq 0$ and $\mathbf{R} > 0$ and all must be symmetric. The time-variant Kalman gain is used, and the reference trajectory is applied within $\mathbf{v}$ instead of $\mathbf{u}$ as seen previously. Both $\mathbf{K}(t)$ and $\mathbf{v}(t)$ are used to find the control input,

$$\mathbf{u} = -\mathbf{K}(t)\mathbf{x} + \mathbf{R}^{-1}\mathbf{B}^T\mathbf{v}(t). \quad (12)$$

The control input equation in a continuous system is an affine state feedback due to the additional term added to the linear state feedback.

3.4 Reinforcement Learning (RL)

RL is a form of machine learning in which one or more agents learns to complete a task through trial and error. During each episode, which is the series of interactions between an RL agent and the environment, the agent chooses an action A_t , based on input observation o_t . Once the chosen action is executed, the environment transitions to a state s_{t+1} , output and observation $\Omega(o_{t+1} | s_{t+1}, A_t)$, and a reward $r_t = \mathbf{R}(s_t, A_t, s_{t+1})$. Figure 2 gives the schematic overview of the method.

The environment can be comprised of any system, from Atari simulations (Hamilton et al., 2020; Alshiekh et al., 2018) to complex robotics scenarios (Brockman et al., 2016; Fisac et al., 2018; Henderson et al., 2018; Mania et al., 2018; Jang et al., 2019; Bernini et al., 2021). RL is based on the reward hypothesis that all goals can be met with the maximization of cumulative rewards (Silver, 2015), but the values of these rewards must be experimented with for the best solution. In this research, the agent receives the full state as an observation $o_{t+1} \in \mathbf{O}$.

Over all episodes, the solution RL finds is the optimal behavior function for the situation and ICs, $A = \pi^*(s)$, that maximizes the expected cumulative sum of rewards:

$$\mathbb{E}[J] = \mathbb{E}\left[\sum_{t=0}^T \gamma^t r_t\right], \quad (13)$$

where T is a finite-time horizon. When $\gamma = 0$, the solution prioritizes immediate reward and when $\gamma = 1$ the solution prioritizes maximizing the expected sum of future rewards. In this case, the reward functions cover actions such as traveling towards the rendezvous orbit, reaching an unsafe velocity, crashing, and other actions considered beneficial or detrimental to the mission. Training of the RL agent concluded once the cumulative change in velocity ΔV and rendezvous time remained stagnant after a series of slightly differing values for each reward function, and the best model's convergence was utilized. This research uses a Proximal Policy Optimization (PPO) algorithm (Schulman et al., 2017).

3.5 Tuning of the Controllers

The only values being changed for each case are the ICs in the format $\mathbf{x}_o = [x_o, y_o, \dot{x}_o, \dot{y}_o]^T$, and the matrices $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{P}$ for the linear quadratic controllers. Bryson's Rule was utilized to tune the weight matrices of the LQR with the objective of limiting control and velocity by designating a maximum value of the states and control. In the application under consideration, $\mathbf{Q} \in \mathbb{R}^{4 \times 4}$ and $\mathbf{R} \in \mathbb{R}^{2 \times 2}$ are given below:

$$\mathbf{Q} = \text{diag}[1/Q_x^2, 1/Q_y^2, 1/Q_{\dot{x}}^2, 1/Q_{\dot{y}}^2] \quad (14)$$

$$\mathbf{R} = \text{diag}[1/R_x^2, 1/R_y^2] \quad (15)$$

A representative example of tuning values chosen using Bryson's Rule is the case in which the maximum distance from the origin is 200 m, and the maximum desired velocity outputs and control inputs are 0.5 m/s and 1 N, respectively. This leads to an initial designation of $Q_{x,y} = 200$, $Q_{\dot{x},\dot{y}} = 0.5$, and $R_{x,y} = 1.0$.

LQI is based around reducing the error calculated from another control method's solution. For this particular case, the results of an LQR are considered for use in Eq. 8. The main difference between LQR and LQI is the added element of complexity in the latter with the usage of $\boldsymbol{\xi}$ for terms $\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{Q}}, \bar{\mathbf{S}}$, and $\bar{\mathbf{K}}$ being resized. An additional scalar multiplier term is added to both the -C section of $\bar{\mathbf{A}}$ and the Identity matrix used in Eq. 9 as an additional tuning element, decided to be 0.6 for this problem. Bryson's rule is not applicable for tuning LQI, but the $\mathbf{Q}$ matrix is tuned in a very similar way to the previous method, with the addition of two tunable values for $\boldsymbol{\xi}$ to make the diagonal matrix shown below to tune position, velocity, and error:

$$\bar{\mathbf{Q}} = \text{diag}[Q_x, Q_y, Q_{\dot{x}}, Q_{\dot{y}}, Q_z, Q_{\xi_x}, Q_{\xi_y}] \quad (16)$$

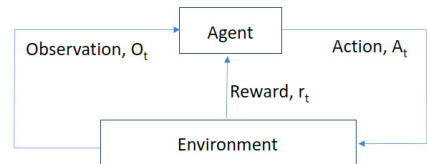


Fig. 2. Agent and Environment Interaction in RL.

The LQT controller is tracking the reference trajectory as a whole, rather than limiting the error of a moving trajectory point. The Q used for this isn't the same size as LQR and LQI and instead reduces to a 2×2 , focused primarily on the position. R is maintained as a 2×2 matrix, regardless of the changes of Q 's size. P is chosen to be very small for every case, which is valid for $P \geq 0$. Another LQT-only tuning measure is that the output can also be tuned with changing the final time T , but the scenario limits defined this as two orbits. The values used for accurately weighting Q , R , P , and in some cases T are decided upon by trial and error, similarly to LQI.

During training, the RL ICs start at a random point in space and end somewhere along a randomly oriented eNMT, so that the RL agent can learn generalized behavior. The reward function used in this research penalizes high relative velocity and acceleration (ΔV), which corresponds to high fuel use. Additional rewards are given for avoiding crashing and reaching the parking orbit.

3.6 Evaluation

The performance of each controller is compared based on cumulative change in velocity (ΔV) and the time used for the deputy to reach $\mathbf{r}_t$. The change in velocity was calculated as

$$\Delta V = \sum \frac{|u_x| + |u_y|}{m} * \Delta t. \quad (17)$$

With discrete control being utilized for RL with a zero order hold on control for a specific time step, the continuous solutions for LQR, LQI, and LQT are discretized for a Δt of 1.0 for comparability. The competing objectives are to minimize total ΔV as well as total time to rendezvous.

Table 1 lists the sets of ICs for the four cases used to compare controller results.

Table 1. Initial Positions

Case No.	x[m]	y[m]	$\dot{x}$ [m/s]	$\dot{y}$ [m/s]
Case 1	0	100	-0.1	0.1
Case 2	-100	200	-0.25	-0.1
Case 3	200	300	0.2	-0.2
Case 4	-300	400	0.1	0.25

4. SIMULATION RESULTS

This section compares results from each controller for each of four different IC cases. Table 3 summarizes the ΔV in m/s obtained for each of the simulation scenarios.

Table 2. ΔV Results [m/s]

Method	Case 1	Case 2	Case 3	Case 4
LQR	3.306	2.072	1.626	3.907
LQI	2.747	5.566	1.539	4.971
LQT	0.633	1.262	0.603	1.423
RL	3.774	4.763	4.053	6.276

Table 3. Rendezvous Time [s]

Method	Case 1	Case 2	Case 3	Case 4
LQR	913	2993	3760	11226
LQI	10230	10373	11195	10996
LQT	1959	5862	5404	11308
RL	2847	5331	4904	10186

4.1 Case 1

The LQR shows the highest peak in instantaneous control at the very start. The RL agent exhibits slight chatter and ends with the highest cumulative ΔV of 3.774 N, followed by LQR solution ΔV of 3.306 N. The next highest control belongs to LQI, with a value of 2.747 N. The ΔV decreases substantially for LQT, at 0.633 N. The LQR and RL maximum velocities are tied at just about 0.4 m/s, as shown in Figures 3-4.

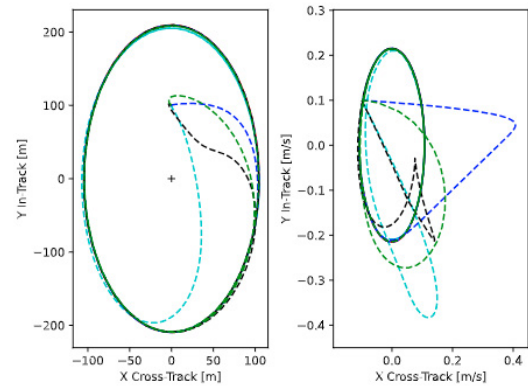


Fig. 3. Case 1: Position and Velocity Outputs.

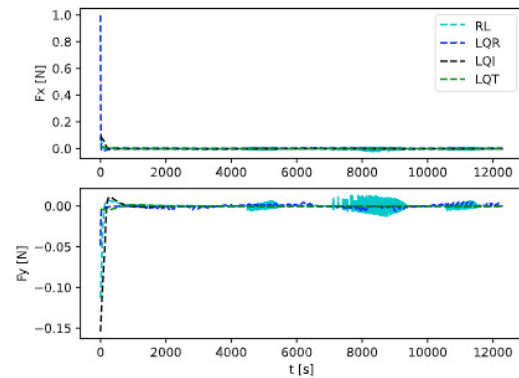


Fig. 4. Case 1: Control Inputs.

4.2 Case 2

In Case 2, the RL agent violates the maximum velocity limit of 0.5 m in the y -direction, reaching a peak of around 0.6 m/s. This violation is signified by the intersection of the RL agent with the gray tinted region in Fig. 5. The LQR controller has the a high instantaneous control at the start. The LQI controller has the highest ΔV at 5.566 N, followed closely behind by RL at 4.763 N. The next highest value is 2.072 N, belonging to the LQR, and the lowest

controller is LQT at 1.262 N. This particular case results in the RL Agent skimming by the origin, which would be a risky trajectory.

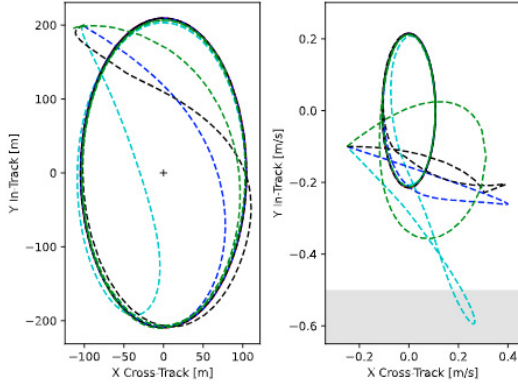


Fig. 5. Case 2: Position and Velocity Outputs.

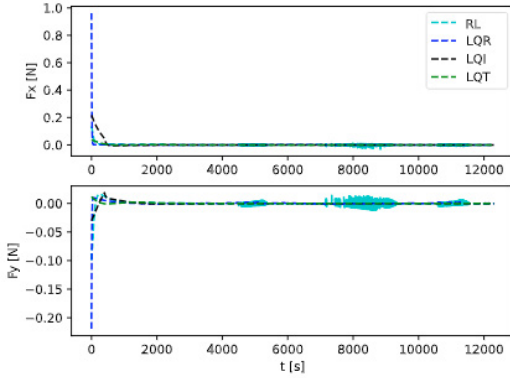


Fig. 6. Case 2: Control Inputs.

4.3 Case 3

The LQI output shows a close pass of the trajectory to the origin, where the chief satellite is positioned. Although the ΔV is not the highest at 1.539 N, it is not enough of a benefit to declare this particular controller the optimal selection. Alongside this disadvantage is the very near miss of the LQI velocity limit that shows that, while it doesn't exceed the safe bounds, it will not be the best choice of trajectory. Interestingly enough, the other three controllers have very similar trajectories coupled with varying ΔV outputs. The ΔV values of Case 3 are of the same order of magnitude as Case 1, as shown in Table 3. Regardless of the negative reward, the RL agent velocity extends past the bounds of velocity, in Fig. 7 but also has a ΔV of 4.053 N.

4.4 Case 4

This particular set of ICs results in the highest ΔV s for most of the control methods, with the RL controller claiming the highest value of 6.276 N, followed by the LQI output with 4.971 N. This is trailed by the LQR and LQT values of ΔV , 3.907 N and 1.423 N. All four controllers are within the time limit of two orbits and the three modern controllers stay within the limit for velocity.

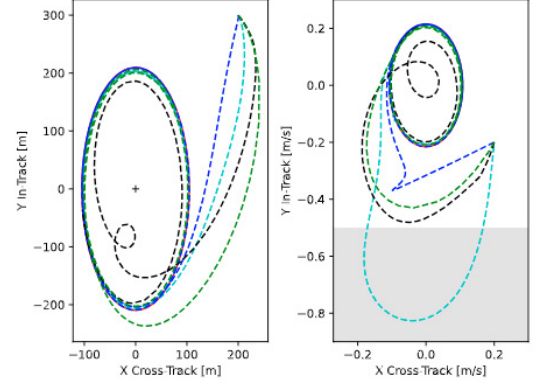


Fig. 7. Case 3: Position and Velocity Outputs.

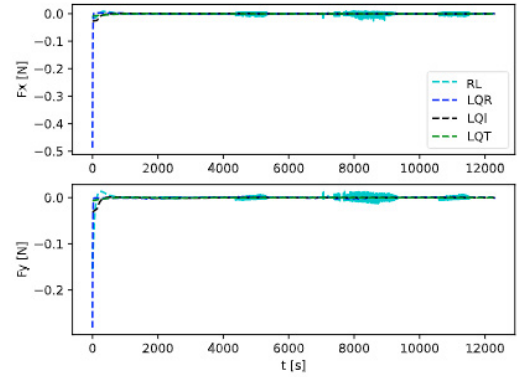


Fig. 8. Case 3: Control Inputs.

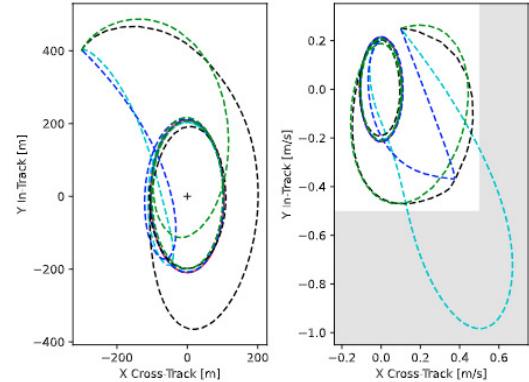


Fig. 9. Case 4: Position and Velocity Outputs.

5. CONCLUSION

In this paper, modern optimal control architectures were compared with RL on a problem involving safe egress to a distant eNMT. The simulation configurations covered LQR, LQI, LQT, and RL using the Clohessy-Wiltshire models for spacecraft dynamics in 2D. We have compared controller performances with respect to time and largest cumulative variation in velocity on four different cases, each characterized by different ICs.

LQR is the simplest modern control method discussed and has the basis of controlling the output to maintain a desired constant value, usually zero, for x . By including

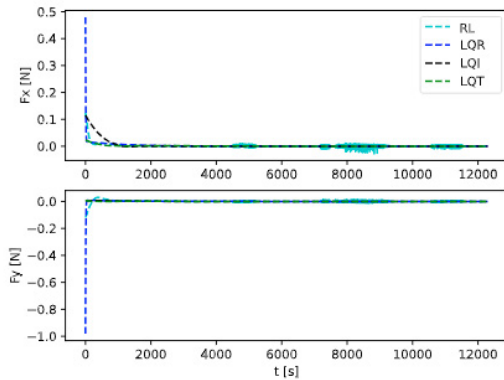


Fig. 10. Case 4: Control Inputs.

the reference trajectory in the calculation of u , LQR is converted to maintaining the value of zero of $(x-r)$ instead and following the desired eNMT orbit. This edit works somewhat well for the purpose and keeps within the safety limits chosen, but the total ΔV is higher than other control methods set within the same problem.

LQI is excellent for reducing steady state error to zero for a constant value rather than following a trajectory. Adding the integral of the error greatly decreases the steady state error and makes any sort of disturbance, force, or model uncertainty insignificant. The positive outcome from converting the LQR to follow a time-based trajectory instead of simply reducing the x to the desired value is applied to LQI as well, but the outcome has greater ΔV and rendezvous time. The integral action and disturbance rejection of LQI increases the robustness of the model and would be considered a major benefit, but is not displayed by the chosen problem.

LQT is used for more complex trajectory tracking in particular, rather than maintaining a constant value by reducing x to the desired constant value. Regardless of the simple trajectory instead of a complex one, the problem statement of falling into a specific eNMT orbit is best done by LQT out of all modern control methods. However, the strict time limit T required the spacecraft navigation solution to end, rather than $t \rightarrow \infty$ as LQI and LQR can.

The RL agent's output is comparable to the modern control methods, as shown by the ΔV and rendezvous time comparisons. A better trained model may result in a better maneuvering, but limited training time and experience created slight issues that ultimately affected the final results, such as chattering and exceeding the desired safety threshold. Even so, there is potential for the future of RL in spacecraft maneuvering with a more adept model and environment used, as well as more time to train to achieve optimal results that stay within the velocity limit of the structural damage threshold.

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